

ZEYU (ALBAN) LI, Ph.D.

Nucleic Acid Process & Formulation Scientist | Ithaca, NY | 626-826-5327 | zl788@cornell.edu | zeyuli.net | linkedin.com/in/albanli

Nucleic acid process & formulation scientist with full-cycle R&D experience in polymer-based delivery formulation, hydrogel and bioconjugation chemistry, continuous-flow biomanufacturing, and downstream purification. Research spans biodegradable PLGA microparticles for synthetic DNA encapsulation, DNA-immobilized hydrogel cartridges sustaining 100-day continuous mRNA production, perfluorocarbon-based high-capacity nucleic acid capture media, and stimuli-responsive biopolymer hydrogel networks. Process discipline includes systematic DOE-driven formulation optimization and positive-pressure cleanroom operation. 6 peer-reviewed publications, 2 patents.

SKILLS

Molecular & Bioengineering: PCR; qPCR; In vitro transcription (IVT); NHS-amine bioconjugation; Enzymatic assays and workflows; Electrophoresis (PAGE, agarose); DNA/RNA extraction & purification; Bead-based nucleic acid purification; Nucleic acid quantification (NanoDrop, UV-Vis); LSPR-based binding assays; Cleanroom protocols (positive-pressure); Liquid chromatography; Fluorometry (Qubit, plate reader).

Characterization: SEM; DLS; Optical microscopy; UV-Vis spectroscopy; Fluorescence microscopy; Oscillatory rheometry; Gravimetric swelling; Tensile testing (Instron); EDS; Contact-angle measurement.

Fabrication & Processing: Microfluidic chip fabrication (soft lithography, PDMS); Continuous-flow bioprocess design; Double-emulsion microencapsulation (PLGA microparticles); Hydrogel formulation; UV photocrosslinking; Surface functionalization; Freeze-drying; 3D printing (DIW, SLA, FDM); Electrospinning; Photolithography; Thermal imprinting; Mold casting; Spin coating; Two-photon lithography; Etching; Nanopatterning; Roll-to-roll processing.

Programming & Data: Statistical DOE; Python; Data analysis & visualization (pandas, NumPy, seaborn); ImageJ; Machine learning (TensorFlow, PyTorch); Git/GitHub; AI-assisted data analysis and research workflows.

Design & Visualization: Scientific figure production (Adobe Illustrator, Photoshop, InDesign); CAD; SketchUp; Blender; KeyShot.

EDUCATION

Cornell University

Biological and Environmental Engineering | Concentration: **Biological Engineering**, Materials Science, Bioenvironmental Engineering

Ithaca, NY

Ph.D. | May 2026 | M.S. | Aug 2024 | M.Eng. | May 2020

Hong Kong Baptist University

Chemistry (Major) | Computer Science (Minor)

Kowloon, Hong Kong

B.Sc. (Hons) | Nov 2019

EXPERIENCE

Graduate Research & Teaching Assistant

Feb 2021 – May 2026

DNA Materials Lab, Advisor: Prof. Dan Luo | Cornell University, Ithaca, NY

- **Formulated biodegradable PLGA polymeric microparticles for synthetic DNA encapsulation and environmental tracing** via double-emulsion solvent evaporation; produced at gram scale under positive-pressure cleanroom conditions with SEM-characterized particle morphology. Validated platform performance in a DoD-funded field deployment, with encapsulated cargo remaining intact and qPCR-detectable at 7 km downstream from sub-milligram release across an 11 km² area. Demonstrating payload protection and sustained release under complex-matrix conditions. (First-author paper, *ES&T*, 2025.)
- **Engineered a continuous-flow mRNA biomanufacturing platform (NucleoMesh) via NHS-amine bioconjugation of DNA templates onto a PEG hydrogel-mesh cartridge**; sustained 100-day steady-state mRNA production at milligram-scale yield, with 7-day demonstrated performance across 6 diverse RNA species (mRNAs, structural and regulatory RNAs). Systematic optimization of hydrogel formulation and flow parameters delivered sustained template and enzyme productivity vs. rapid saturation in batch IVT, with significantly lower dsRNA impurity maintained throughout the run. (Second-author paper, *Nature Sustainability*, in peer review.)
- **Developed a pod-based continuous-flow nucleic acid purification platform using perfluorocarbon as a high-capacity capture medium**; achieved 41.73 µg/µL binding capacity (800× silica-based conventional methods),

translating to ~209 mg DNA loading per pod with over 90% capture efficiency. Reduced consumables cost by ~91% versus ethanol precipitation, enabling scalable downstream purification for qPCR and sequencing workflows.

- **Developed stimuli-responsive DNA hydrogels via thermal, photochemical (UV-reversible psoralen), and ionic (Al^{3+}) crosslinking, spanning thermoreversible to irreversible networks from a single biocompatible biopolymer feedstock.** Characterized 182 formulations across 6 variables via oscillatory rheometry, tensile testing, and gravimetric swelling. (First-author paper, *ACADIA*, 2024).
- **Co-instructed biomaterials classes and mentored 3 undergraduate researchers;** designed a 6-lecture computational tooling module integrating AI-assisted analysis into biomaterials workflows and organized a Biomaterials Hackathon. Teaching assistant across 7 engineering courses total. *Cornell CALS Outstanding Teaching Assistant Award*.

Senior Research Assistant & Undergraduate Research Assistant

Jun 2017 - Jan 2021

Microfabrication & Surface Materials Lab, Supervisor: Prof. Kangning Ren | Hong Kong Baptist University

- **Engineered monolithic perfluoropolymer superhydrophobic surfaces** via high-temperature soft imprinting and thermal fusion with master molds from photolithography, two-photon lithography, and etching; validated topology-specific superwettability and ultradurability across abrasion, compression, corrosive vapors, UV aging, thermal treatment, and 100,000+ folding cycles. (Third-author paper, *The Innovation* IF = 33.2; US Patent 11,839,998).
- **Designed and fabricated an automated weather-environment simulation chamber** integrating temperature and humidity control, supercooled droplet generation, adjustable droplet parameters, sample-stage cooling, and high-speed monitoring; incorporated an electrostatic surface-charging module to quantify voltage-dependent droplet behavior on superhydrophobic surfaces. Demonstrated ~25% reduction in droplet contact time under charged icing conditions. *Best Undergraduate Thesis Award*.

Research Exchange Trainee

Jun 2018 – Aug 2018

Molecular Basis of Disease Program | Georgia State University, Atlanta, GA

- **Characterized protein-DNA binding kinetics using LSPR** (localized surface plasmon resonance) assays, a label-free real-time technique analogous to SPR. Quantified sequence-specific binding affinities of transcription factor PU.1 to DNA targets. *Best Poster Presentation Award*.

PUBLICATIONS

- Wang, D., [Li, Z.](#), Li, J., Han, Y., Sun, T., Li, F., Liu, P. "A Chip Reactor for Perpetual Nucleic Acid Production and On-chip Information Processing." *Nature Sustainability*, in peer review (2026).
- [Li, Z.](#), Ramón, C. L., Koeberle, A., et al. "Tracing Environmental DNA Transport in Large Lakes with Synthetic DNA Microparticles and Hydrodynamic Modeling" *Environmental Science & Technology* (IF = 12.4), (2025). [DOI](#)
- He, C., [Li, Z. \(co-first authors\)](#), Wang, L. X., et al. "PolyTile 4.0: Self-healing Ceramic Tiles" *ACADIA* (2024). [DOI](#)
- Li, W., Chan, C. W., [Li, Z.](#), et al. "All-perfluoropolymer, nonlinear stability-assisted monolithic surface combines topology-specific superwettability with ultradurability." *The Innovation* (IF = 33.2), 4(2), 100299, (2023). [DOI](#)
- [Li, Z.](#), Liu, Z., Yang, X., Luo, D. "What the Records Don't Carry: A Position on Researcher-AI Co-Adaptation in Exploratory Laboratory Research." Under review at **ICML Workshop on AI for Science**, (2026).
- [Li, Z.](#) "Apprenticeship Without Proximity: An Agent Skills Architecture for Unwritten Experimental Judgment." Under review at **ACM CAIS Workshop on Agent Skills**, (2026).

PATENTS

- Ren, K., Wu, H., Wang, Z., Yao, S., Ong, B., Li, W., [Li, Z.](#), Sun, H., & Chan, C.W. "Crack engineering as a new route for the construction of arbitrary hierarchical architectures." US Patent 11,839,998, (2023).
- Li, Q., [Li, Z.](#), & Lin, Z. "Reactor and method of spiral propulsion biomass continuous thermal cracking." Chinese Patent 201711214139.6 (2017).